Jayakrishna M

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Senior Backend Engineer | Java| Python | AI/ML | Cloud | Distributed Systems

Senior Backend Engineer with 10+ years of experience building high-throughput, low-latency financial systems. Specialized in event-driven architecture, microservices, and cloud-native platforms, with strong expertise in transaction processing, system reliability, and scalability. Proven track record of designing fault-tolerant systems handling large-scale concurrent workloads with high availability.

Core Expertise

- Languages: Java (8+), Python, SQL
- Frameworks: Spring Boot, Spring, Hibernate
- Architecture: Microservices, Event-Driven, CQRS, REST, gRPC
- Messaging: Apache Kafka (high-volume event processing)
- Cloud: AWS (EC2, S3, RDS, Lambda)
- Containers: Docker, Kubernetes, OpenShift
- Databases: PostgreSQL, Cassandra, MongoDB, MySQL
- Caching: Redis
- CI/CD: Jenkins, Tekton Pipelines
- Security: OAuth2

System Scale & Impact

- Designed systems handling high-volume concurrent transactions with 99.9%+ availability
- Improved API response latency by 30–50% using caching and query optimization
- Built event-driven pipelines processing millions of events daily
- Enabled horizontal scalability supporting high traffic financial workloads

Key Engineering Achievements

- Designed scalable Spring Boot microservices handling high-volume concurrent requests in production
- Built event-driven systems using Kafka (consumer-based processing) to improve scalability and decouple services
- Integrated Redis caching to reduce database load and improve response times
- Optimized SQL queries using indexing and pagination, reducing response time by ~40% and improving API throughput.
- Developed fault-tolerant systems on AWS using Auto Scaling, Load Balancers, and resilient design patterns

- Implemented retry mechanisms and circuit breaker patterns to improve system reliability
- Designed stateless services enabling horizontal scaling and high availability
- Designed HLD and LLD for Kafka-driven microservices, including event schemas, topic partitioning strategy, and idempotent processing, ensuring reliable and scalable event-driven architecture.

Professional Experience

IBM — Advisory System Analyst (Backend & Cloud)

Duration: [2021- Till date]

- Designed and developed cloud-native microservices for financial transaction systems, ensuring high availability and low latency (<200ms)
- Built Kafka-based event-driven architecture to process large-scale asynchronous workloads, improving system scalability and decoupling services
- Implemented resilient design patterns (retry, circuit breaker) to enhance fault tolerance and system stability
- Developed and maintained CI/CD pipelines using Tekton and OpenShift, reducing deployment time and improving release reliability
- Optimized backend performance using caching (Redis) and database tuning, improving response times by ~40%
- Contributed to SLA-critical financial systems, ensuring uptime and reliability requirements

Accenture — Application Development Senior Analyst

Duration: [August 2020 – Dec 2021]

- Developed enterprise-grade backend systems using Spring Boot and REST APIs
- Designed scalable distributed systems, improving system throughput and performance
- Optimized SQL queries and indexing strategies, reducing database load and latency
- Collaborated in building high-availability systems supporting critical business workflows

Capgemini — Associate Java Consultant

Duration: [May 2018- June 2020]

- Built backend services and contributed to large-scale system integration projects
- Developed REST APIs and supported enterprise application development
- Worked on improving system performance and reliability

AJR Info Systems — Java Trainee

Duration: [Aug 2015- Sep 2017]

- Developed foundational expertise in Java, Spring, and web services
- Contributed to backend development and debugging tasks

Key Projects

Financial Products Platform (PPF, SSY, NPS)

- Designed and developed scalable microservices for government-backed financial products
- Built secure transaction processing systems handling high concurrency
- Implemented Kafka-based asynchronous processing, improving throughput and reliability
- Reduced latency by 30% using Redis caching and optimized queries
- Deployed on OpenShift ensuring high availability and scalability

CICS Modernization

- Modernized legacy mainframe systems by exposing CICS services as REST APIs
- Improved system interoperability between legacy and modern distributed systems

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Low-Code Integration Platform

- Built backend services enabling third-party API integrations
- Implemented secure authentication using OAuth2
- Designed scalable architecture supporting multiple integrations

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Tech Platform (GCP Deployment)

- Developed REST APIs using Spring Boot
- Designed scalable database architecture on GCP
- Led deployment and improved application performance

System Design & Engineering Expertise

- High-scale distributed systems design (scalability, availability, reliability)
- CAP theorem, consistency models (ACID vs BASE)
- Event-driven architecture and microservices patterns
- Fault tolerance (retries, idempotency, circuit breakers)
- Caching strategies and performance optimization
- Observability: logging, monitoring, alerting

Certifications

- IBM Certified Technical Advocate – Cloud V3
- Red Hat Certified Specialist in Containers
- IBM Cloud for Financial Services

Education B.Tech in Computer Science and Engineering